

6	D	The equation is for the conservation of linear momentum in the vertical direction.
7	A	Pressure at the boundary between the oil and water in the left arm is the same as the pressure at the same height due to water on the right arm. $30.0\rho_{oil}g = (30.0 - x)\rho_{water}g \Rightarrow x = 4.5 \text{ cm}$
8	C	By conservation of energy, work done by the force $W =$ gain in kinetic energy ΔK